

DATA612 Decision Analytics
Group Assignment #1 – Landhills Winery
In-Class Session: March 11th, 2026
Report Submission: March 16th, 2026

Answer the following questions based on the Landhills Winery case study.

Question 1. (Base Model, 80%) Put yourself in shoes of Richard Thornton and develop a business plan to be submitted to the board by (at the least) answering the following questions.

1. How much of each wine should be produced to maximize net profits?
2. Which grape type is the most valuable for them? What would you recommend to the senior management in terms of seeking more supply of grapes?
3. There are several constraints imposed by the government and the company's internal policies. Identify a constraint that you would have preferred to change by 1% (up or down) if you had the chance and explain your reasoning and the value that it would bring.

Question 2. (Extension 1, 20%) Suppose that the grower of Merlot 2010 offers a quantity discount so that each bottle is offered for \$1.1 if the order size is greater than 150,000. Modify your base model to accommodate this new pricing scheme. What is the new optimal production plan? How does it compare to the original plan? The mathematical formulation (with binary variable) required to solve this problem is available on D2L. Only mathematical programming approach will be accepted.

Guidelines and Rubric:

1. In-Class Session (70%): I am expecting you to finish at least Question 1 in-class and present the results as a business case by impersonating Richard Thornton. You will be asked to submit your work (Excel or Python model) at the end of the class. Each team will present their work briefly at the end of the class. If you have a running model that finds the optimal solution to Question 1 by 7:15 pm and present the results satisfactorily, then you will secure 80% of this assignment.
2. Report (30%): Your report should include all the answers to the questions above. The report should be in a business format that can be submitted to the executive board of Landhills Winery by Richard Thornton.
3. Up to 10% can be deducted for poor style and formatting of the report.
4. In addition to the submission at the end of the in-class session, you will submit your model and report by the second deadline.
5. It is academic misconduct to communicate with anyone else other than your own group members about the assignment. Use of any AI tool is also not permitted for this assignment. If you use any electronic or physical references or tools in solving any of the questions or writing your report, you must provide reference.